

1 - Método da Bissecção

2 -

$$2) \begin{cases} 3x_1 - 4x_2 + x_3 = 9 \\ x_1 + 2x_2 + 2x_3 = 3 \\ 4x_1 + 0x_2 - 3x_3 = -2 \end{cases}$$

$$\left[\begin{array}{ccc|c} 3 & -4 & 1 & 9 \\ 1 & 2 & 2 & 3 \\ 4 & 0 & -3 & -2 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 3 & -4 & 1 & 9 \\ 0 & 10/3 & 5/3 & 0 \\ 4 & 0 & -3 & -2 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 3 & -4 & 1 & 9 \\ 0 & 10/3 & 5/3 & 0 \\ 0 & 16/3 & -13/3 & -14 \end{array} \right]$$

$L_2 \rightarrow L_2 - \frac{1}{3}L_1$ $L_3 \rightarrow L_3 - \frac{4}{3}L_1$ $L_3 \rightarrow L_3 + \frac{4}{3}L_1$ $L_3 \rightarrow L_3 - \frac{8}{5}L_2$

$$\begin{aligned} 3x_1 - 4x_2 + x_3 &= 9 \\ \frac{10}{3}x_2 + \frac{5}{3}x_3 &= 0 \\ -\frac{21}{3}x_3 &= -14 \end{aligned}$$

$$x_3 = -14 \cdot \frac{3}{-21}$$

$$x_3 = \frac{42}{21}$$

$$\boxed{x_3 = 2} //$$

$$\begin{aligned} \frac{10}{3}x_2 + \frac{5}{3}x_3 &= 0 \\ \frac{10}{3}x_2 + \frac{5}{3} \cdot 2 &= 0 \\ \frac{10}{3}x_2 &= -\frac{10}{3} \\ x_2 &= -\frac{10}{3} \cdot \frac{3}{10} \\ \boxed{x_2 = -1} // \end{aligned}$$

$$\begin{aligned} 3x_1 - 4x_2 + x_3 &= 9 \\ 3x_1 - 4(-1) + 1(2) &= 9 \\ 3x_1 &= 9 - 4 - 2 \\ 3x_1 &= 3 \\ \boxed{x_1 = 1} \end{aligned}$$

Verificação

$$\begin{aligned} 3 \cdot 1 - 4(-1) + 2 &= 3 + 4 + 2 = 9 \\ 1 + 2(-1) + 2(2) &= 1 - 2 + 4 = 3 \\ 4 \cdot 1 - 3 \cdot 2 &= -2 \end{aligned}$$

3 -

$$^3) \{ (1, 3), (4, 18) \}$$

x_K	1	3
y_K	4	18

$$n+1=2 \Rightarrow n=1$$

~~$$P_1(x) = 2x + 21x$$~~

$$P_1(x) = \sum_{K=0}^1 y_K L_K(x) = y_0 L_0(x) + y_1 L_1(x) = \boxed{4L_0(x) + 18L_1(x)} \text{ eq I //$$

$$L_K(x) = \prod_{i=0, i \neq K}^1 \frac{(x - x_i)}{(x_0 - x_i)}, \quad K=0, 1$$

* Se $K=0$

$$L_0(x) = \prod_{i=1, i \neq 0}^1 \frac{(x - x_i)}{(x_0 - x_i)} = \frac{(x - 3)}{(1 - 3)} = \boxed{\frac{x-3}{-2}} //$$

* Se $K=1$

$$L_1(x) = \prod_{i=0, i \neq 1}^1 \frac{(x - x_i)}{(x_0 - x_i)} = \frac{(x - 1)}{(3 - 1)} = \boxed{\frac{x-1}{2}} //$$

Voltando em eq I

$$P_1(x) = 4L_0(x) + 18L_1(x) = 4 \cdot \left(\frac{x-3}{-2} \right) + 18 \cdot \left(\frac{x-1}{2} \right) = \frac{4x-12}{-2} + \frac{18x-18}{2} = -2x+6+9x-9 = -2x+9x+6-9 = \boxed{7x-3} //$$

$$x=1$$

$$7 \cdot (1) - 3 = \boxed{4}$$

$$x=3$$

$$7 \cdot (3) - 3 = \boxed{18}$$

4 -

$$4) f(x) = x^2, [0, 2], n = 4$$

$$x_0 = 0, x_1 = \frac{1}{2}, x_2 = 1, x_3 = \frac{3}{2}, x_4 = 2$$

$$\int_0^2 x^2 dx = \sum_{i=0}^3 \frac{(x_{i+1} - x_i) \cdot (f(x_{i+1}) + f(x_i))}{2}$$

Para $i = 0$

$$\frac{(\frac{1}{2} - 0) \cdot ((\frac{1}{2})^2 + 0^2)}{2} = \frac{(\frac{1}{2}) \cdot (\frac{1}{4})}{2} = \frac{\frac{1}{8}}{2} = \frac{1}{16} //$$

Para $i = 1$

$$\frac{(1 - \frac{1}{2}) \cdot (1^2 + (\frac{1}{2})^2)}{2} = \frac{(\frac{1}{2}) \cdot (\frac{5}{4})}{2} = \frac{\frac{5}{8}}{2} = \frac{5}{16} //$$

Para $i = 2$

$$\frac{(\frac{3}{2} - 1) \cdot ((\frac{3}{2})^2 + 1^2)}{2} = \frac{(\frac{1}{2}) \cdot (\frac{13}{4})}{2} = \frac{\frac{13}{8}}{2} = \frac{13}{16} //$$

Para $i = 3$

$$\frac{(2 - \frac{3}{2}) \cdot (2^2 + (\frac{3}{2})^2)}{2} = \frac{(\frac{1}{2}) \cdot (\frac{25}{4})}{2} = \frac{\frac{25}{8}}{2} = \frac{25}{16} //$$

$$\int_0^2 x^2 dx = \frac{1}{16} + \frac{5}{16} + \frac{13}{16} + \frac{25}{16} = \frac{44}{16} = \frac{11}{4} = 2,75 //$$